

What's growing where, this season

An in-season crop classifier from Sentinel-2 imagery, built over four weeks with Claude Code

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THE QUESTION

Two data sources: CDL for past years, a classifier for this one

COMPLETED YEARS

Read it from USDA CDL

The Cropland Data Layer is a 30 m national crop map, published after the season ends. It is itself a classifier, trained on farm program records, with per-state error rates.

THE SEASON IN PROGRESS

Train our own classifier

Sentinel-2 NDVI and red-edge (NDRE) curves per field, labeled with past CDL, scored against the ground. No public map exists until after harvest.

Issue 427 and its sub-issues

#422 · done

**Independent
labels**

#423 · done

**Zones and label
space**

#407 #448 #451

Classifier, done

#424 · open

**Weekly field
inference**

#425 #419

**Pixel raster,
live overlay**

#410 #411

County rollups

Shipped first: Phase A (#409), the historical overlay from CDL. No model required.

Built before the classifier: the independent evaluation set.

Phase A: historical overlay from CDL

0.35 MB

one county-year as PMTiles, z10 to z13, 196 tiles

-64%

archive size from a 10-pixel sieve before tiling

0 req

network requests to toggle a crop class on the map

MapLibre 5 has no paint expression over raster pixel values, so per-class toggling in the style was not possible. A custom protocol caches tile bytes and rewrites the PNG tRNS chunk per request instead. The PR documents the deviation from the spec.

Independent labels, built before the classifier

Testing on held-out CDL only measures how well you copy CDL. We needed labels CDL never touched, frozen in a hashed, write-once file before any model existed.

Option	What the probe found
NAIP aerial photo interpretation	2023 is Alabama's last flight, fragmented over six dates. No survey from 2024 on has a coincident flight.
Sentinel-2 coverage, 2022	Zero scenes. Collection-1 has a hole; the data sits in a fallback archive. Without the fallback, 2022 would have been dropped.
Windshield survey	Drive the county and record what is standing. Aug 29, Madison County: 94 fields, GPS points to field polygons.

Zone 0110: Alabama Northern Valley

USDA NASS divides each state into Agricultural Statistics Districts, 1 to 13 per state. Zone ID = state FIPS **01** + district **10**. Alabama has six:

Zone	District	Counties
0110	Northern Valley	8
0120	Mountains & Eastern Valley	9
0130	Upper Plains & Piedmont	12
0140	Black Belt	12
0150	Coastal Plains & Gulf Coast	10
0160	Wiregrass	10

Why per zone, not one national model

- **Calendar:** emergence dates and the bare-field prior are the zone's own.
- **Crop mix:** label space and area-share priors come from the zone's CDL.
- **Label noise:** CDL accuracy is published per state.
- **Checkable:** NASS reports acreage per district.

Expected to beat a national model; not yet measured against one.

Segmenting fields from multi-year CDL

1. Stack CDL for 2020 to 2023 at 30 m.
2. Run a 3×3 majority filter on each year to remove single-pixel misclassifications.
3. Keep cropland: pixels cultivated in at least 2 of the 4 years.
4. Key every pixel by its crop sequence across the years. Connected pixels with the same sequence form one segment.
5. Erode the 1-pixel boundary ring; keep segments of 50+ pixels (4.5 ha) that are 90%+ one crop in the season being labeled.

TWO NEIGHBORING FIELDS

2023 only: both corn, one segment

corn | corn

2020-2023 sequence: two segments

S C S C

C S C C

184,855

components without the filter; median size 1 pixel

914

training segments with the filter and floors

scikit-learn gradient boosting

```
booster =  
HistGradientBoostingClassifier(  
    max_iter=300,  
    learning_rate=0.06,  
    early_stopping=True,  
    validation_fraction=0.15,  
)  
  
model = CalibratedClassifierCV(  
    booster,  
    method="isotonic",  
    cv=PredefinedSplit(season_fold),  
    ensemble=False,  
)
```

- **Tabular input:** 43 phenology and timing columns per field-day. Trees, no GPU, small enough to load in the inference path.
- **Native NaN handling:** a GDD checkpoint the season has not reached is missing, not zero.
- **Calibrated by season:** isotonic map fit on leave-one-year-out scores; sigmoid if a class has under 25 rows.

SCIKIT-LEARN DOCS

[HistGradientBoostingClassifier](#) · [Histogram-based gradient boosting guide](#)

[CalibratedClassifierCV](#) · [Probability calibration guide](#) · [PredefinedSplit](#)

Training and inference in code

TRAIN · offline, per zone

```
from sat_sentry.croptmap import model, training_data
# one row per segment per weekly grid day
rows = pd.concat([
    training_data.simulate_days(
        samples, curves[y], climate, grid_days,
        year=y)[0]
    for y in range(2020, 2026)])
zone = model.train_zone_model(
    "0110", rows,
    label_space=label_space,
    area_shares=area_shares,
    cdl_vintages=(2020, 2021, 2022, 2023),
    contributing_fips=fips, version="v7")
model.save(zone, storage)
```

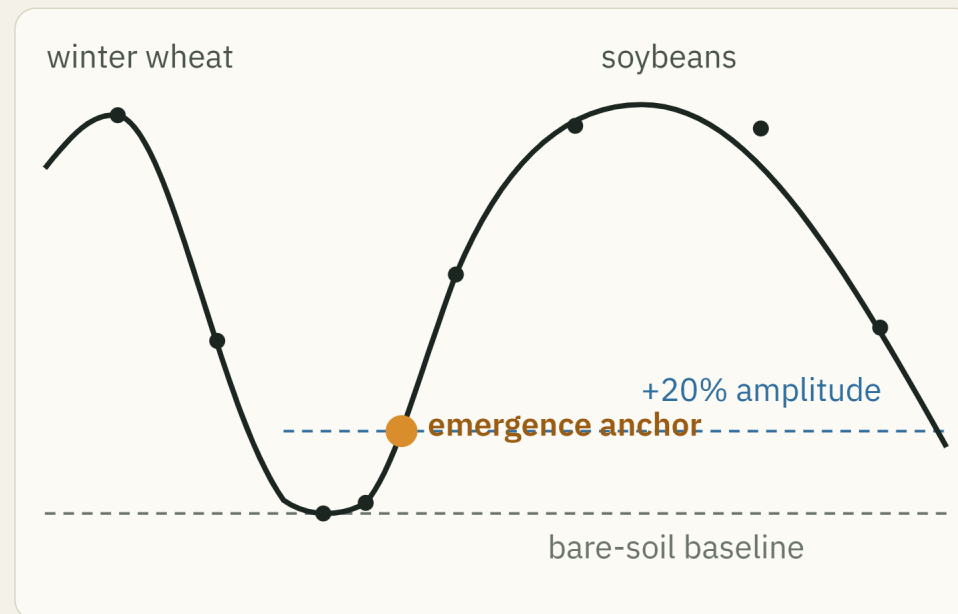
LOAD AND CLASSIFY · weekly

```
from sat_sentry.croptmap import model
zone = model.load(storage, "0110", "v7")
# raises unless feature version is 1.3.0
out = zone.classify_on(
    "2026-07-08",
    curves, # field_id, date, ndvi, ndre
    climate, # date, tmax, tmin (PRISM)
    latitude=34.7)
# one row per field:
# emerged, emergence_date,
# gdd_since_emergence,
# p_corn, p_cotton, p_soybeans,
# source = "model" | "prior"
```

Aligning every field on its own emergence

Planting dates in one county spread over weeks. From a Jan 1 origin, the same crop greens up at different points on the axis. Zero is set per field, from its NDVI curve:

1. Bare-soil baseline: 10th percentile of cloud-masked NDVI. A curve rising less than 0.15 above it has no crop.
2. Find green-up cycles: rises and falls of at least 0.15 NDVI.
3. Take the last cycle that started from bare soil, trough within 20% of amplitude of baseline.
4. Emergence: where NDVI first crosses baseline + 20% of that cycle's amplitude, interpolated between the two clear observations either side.



Growing degree days as the time axis

ONE DAY'S GDD

$$\max((T_{\max} + T_{\min}) / 2 - T_{\text{base}}, 0)$$

A 32 / 20 °C day adds 16 GDD at a 10 °C base.

Daily PRISM temperatures, 4 km grid.

Crops develop by accumulated heat, not by date, so a cool spring and a warm one line up on this axis where they would not on day-of-year.

Crop	Base temp	GDD to emerge	Zone 0110 emergence
Corn	10 °C / 50 °F	61	May 16-30
Soybeans	10 °C / 50 °F	75	Jun 11-25
Cotton	15.6 °C / 60 °F	83	Jun 22-29
Winter wheat	0 °C / 32 °F	67	sown in autumn

The classifier uses **one base, 10 °C, for every crop**, and samples NDVI and NDRE at 13 checkpoints from 100 to 2,800 GDD after emergence.

First model, scored two ways

Gradient-boosted trees on 39 phenology features along a growing-degree-day axis. Trained 2020 to 2022, CDL 2023 held out, curves cut at Aug 29.

Class	Held-out CDL	Ground survey	n	Reading
Corn	0.975	0.953	43	ship
Cotton	0.838	0.939	33	ship
Soybeans	0.894	0.389	18	do not ship

The three classes average 0.760, which clears a 0.75 bar. Classes are gated individually, not on the mean.

0.894

soybeans vs. held-out CDL

0.389

soybeans vs. the ground

The cause was not CDL's soybean labels

Exploration tooling: inspecting the training inputs

A small FastAPI explorer that imports the production feature code, so it shows the same inputs the model trains on. First view: per-class distribution of every feature, with NaN rate and out-of-range count.

1

A whole training year was not NDVI

2

The model learned the calendar from missing values

3

The evaluation never truncated the curves

NDVI is bounded to $[-1, 1]$. The 2022 rows reached 3,591.

2022 came from the fallback Sentinel-2 archive, whose metadata advertises a -0.1 reflectance offset its pixels do not carry. A band goes negative before the ratio and the normalized difference explodes.

Held-out 2023 and the 2026 survey both came from the clean archive, so no accuracy figure could see it.

24%

of training rows were corrupt

97.5%

of 2022 segments had peak NDVI above 1

Single-date fit: the model learned planting date from missing values

Timing leaked through missingness

With curves cut at Aug 29, how far a field got along its own growing-degree axis was the class. Furthest checkpoint 1400: 87% corn. Checkpoint 400: 82% soybeans. The booster splits on NaN natively and found it.

2026 soybeans emerged three weeks early, reached checkpoints training soybeans never had, and **9 of 18 were called corn.**

Evaluation used longer curves than training

The model was fitted on curves ending Aug 29. 90 of 94 survey fields were scored on curves running to Sep 3 to 8.

So 0.389 overstated in-season accuracy.

Day-grid training: every week of every season

Truncate each training curve at every week, mid-April to mid-October

Recompute emergence and features on what was observable that day

Expose timing as explicit columns: day, emergence day, GDD since

Score one season held out at a time, plus a 2 km block holdout

Day	Corn	Cotton	Soybeans	Weighted, by year	Year x block
Jun 24	0.92	0.16	0.94	0.730 ±0.073	0.729
Jul 22	0.96	0.79	0.85	0.878 ±0.014	0.875
Sep 2	0.96	0.87	0.90	0.916 ±0.026	0.912

Prior vs. classification

CLASSIFICATION · EMERGED FIELD

The booster reads this field's own NDVI and NDRE curve and returns calibrated probabilities. Evidence about this field moves the answer.

PRIOR · BARE FIELD

No curve to read yet. Zone's CDL area share per crop × share of that crop not yet emerged that day. Every bare field gets the same answer, marked **source = prior**.

ARGMAX

Return the class with the highest probability and drop the rest. Here every bare field becomes "soybeans": on Jun 24 that called **27 of 33 bare cotton fields** soybeans. So a bare field keeps its probabilities and never sets a season's crop type.

P(crop | bare on Jun 21)

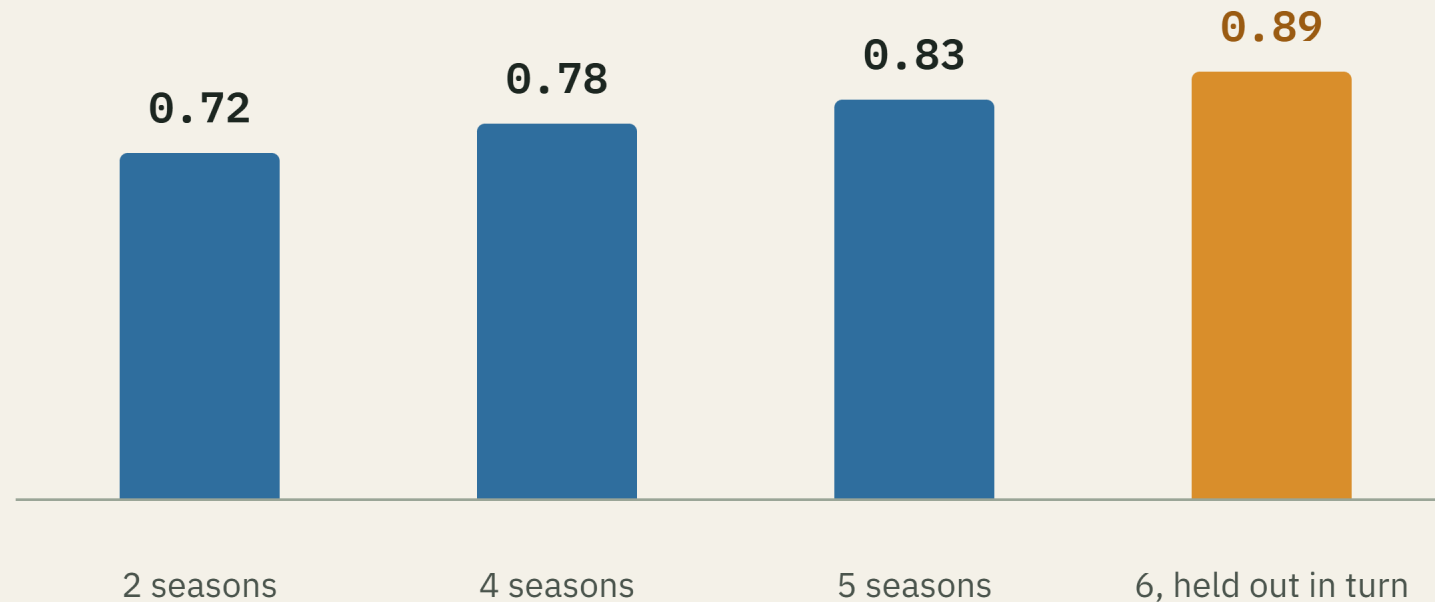
Soybeans		0.61
Cotton		0.36
Corn		0.03

Red-edge transfers to a new year; NDVI alone does not

Feature set (timing-blind, emerged fields)	Held-out 2023	2026 survey
NDVI checkpoints only	0.708	0.585
NDRE checkpoints only	0.716	0.777
Both checkpoint sets	0.893	0.809
All 38 features	0.894	0.840
All 38 plus ten NDRE shape summaries	0.897	0.819

Either alone is 0.71 held-out; together 0.89. The signal is NDRE relative to NDVI. NDRE carries cotton: 0.94 against 0.55 on 2026.

Soybean accuracy vs. number of training seasons



2026 survey soybeans after emergence, by number of training seasons.

Adding 2024, 2025 and a corrected 2022 reused the existing segments instead of re-reading every season's imagery (about 12 hours).

Emergence anchor: v5, v6, v7

v5 · PROBLEM

Anchor before the season's peak. On wheat-then-soybean fields the peak is the wheat: soybeans plateau at 82% anchored.

v6 · FAILED

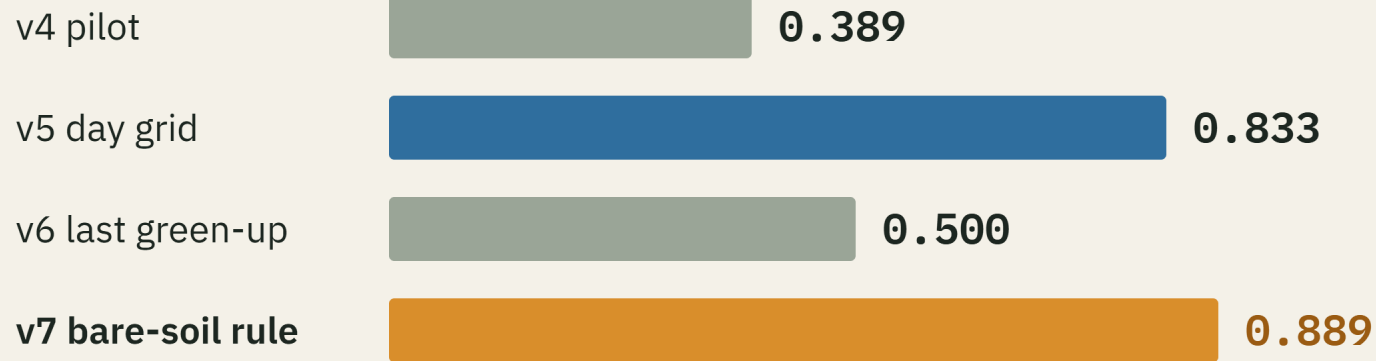
Anchor on the last green-up. One cloud dip in a closed canopy reads as a new crop. Survey soybeans: 0.500.

v7 · SHIPPED

A later cycle counts only if it starts from bare soil: trough within 20% of amplitude above baseline.

Measured cost, left open: October cover crops on harvested corn are real later cycles, and 13.5% of corn anchors move on Oct 17.

Soybeans on the ground survey, version by version



v7, ALL 94 FIELDS, SEP 12

0.947

weighted · corn 0.953 ·
cotton 0.970 · no field
answered by the prior

Held-out seasons, every one in turn: 0.868 on Jul 25 and 0.901 on Sep 5.

Development process

11

Claude Code sessions, Aug
19 to Sep 18

8

merged pull requests

~40k

lines added, 10.6k of
them tests

94

fields in the ground survey

The loop

Discuss in one session, write the spec as a GitHub issue, implement in a fresh session, review and merge the PR. The parent issue holds the current state and every decision with its reason.

The guardrails

AGENTS.md: format, lint, strict mypy and the full test suite before any commit; an IAM review for every new AWS call; docs that stand alone.

TAKEAWAYS

Practices from this project

Freeze the test set before the model

Labels from a source the training data never touched.

Report per class, never an average

0.760 average hid a 0.389.

Look at the inputs

Physical-range checks and NaN rates found what accuracy could not.

Simulate the question users ask

"Classify on day d", scored by held-out year, never a random split.

Have the agent verify spec figures

The spec's 5.8% sorghum figure was wrong.

Persist what you can't regenerate

Ephemeral containers cost a 21-hour cache rebuild.

STILL OPEN

Open work

Issue	What it does
#424	Weekly field inference, the first consumer of the v7 model
#452	Per-zone season read window; also bounds the October re-anchoring
#453	Labels from farmer declarations and roadside observation for other zones
#425 #419	Per-pixel raster and the live in-season overlay

Paused pending customer investigation. After emergence: 0.92 on an unseen season. Before: a prior. **Is that a product?**

FIELD FINDER • ISSUE 427

Questions?

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